

Remote sensing of wave-orbital velocities in the surfzone

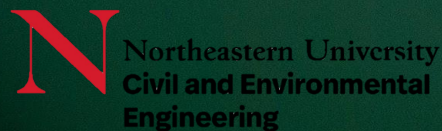
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Introduction

- Can be difficult to measure velocity in the surfzone

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- Cameras can provide accurate surface velocity measurements

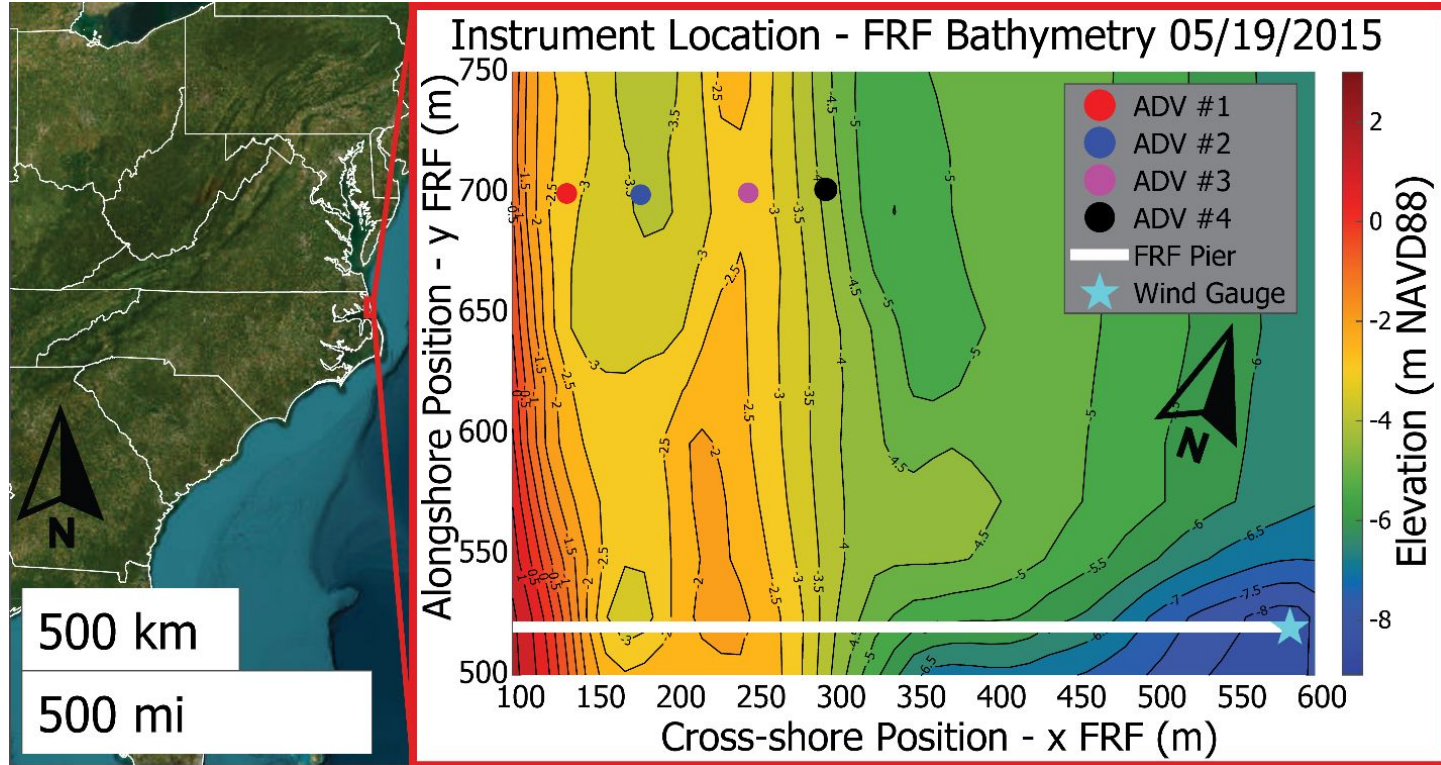
Introduction

- Can be difficult to measure velocity in the surfzone
- Cameras can provide accurate surface velocity measurements
- Linear Wave Theory (LWT) can translate velocity up and down the water column

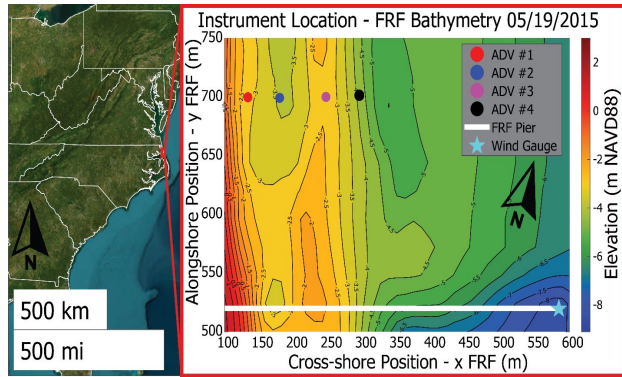
Introduction

- Can be difficult to measure velocity in the surfzone
- Cameras can provide accurate surface velocity measurements
- Linear Wave Theory (LWT) can translate velocity up and down the water column
- **Hypothesis: Consumer video cameras and linear wave theory can be used to estimate wave orbital velocities in the surfzone**

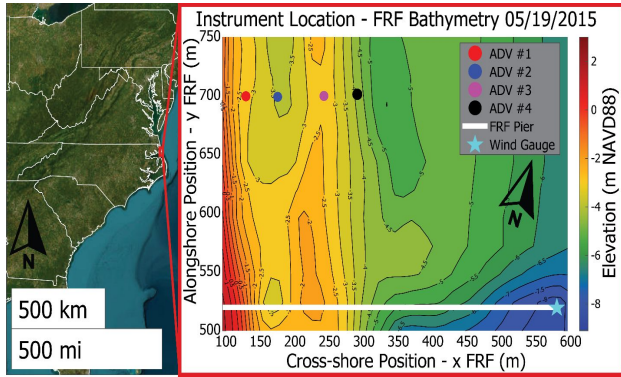
Experimental setup



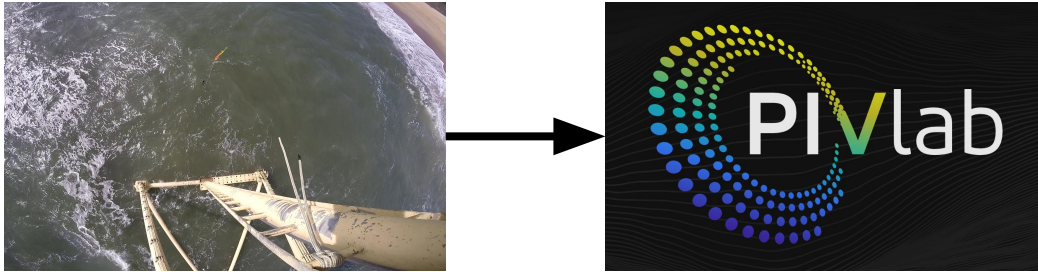
Experimental setup



Experimental setup

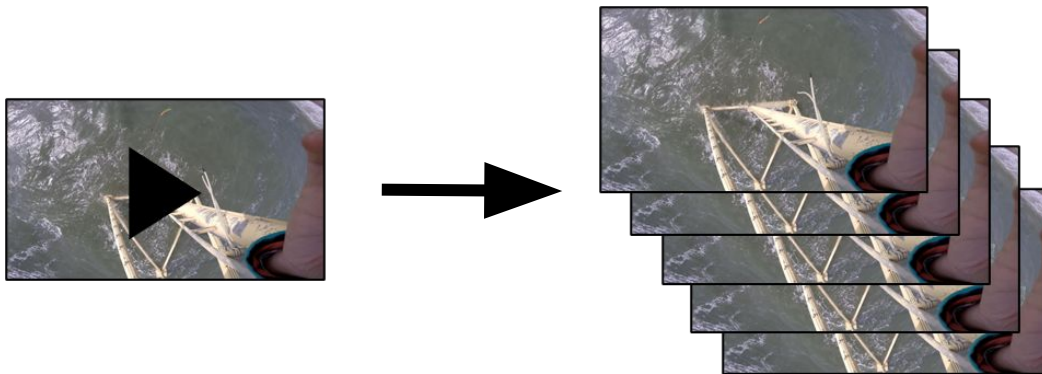


Data Processing- Overview



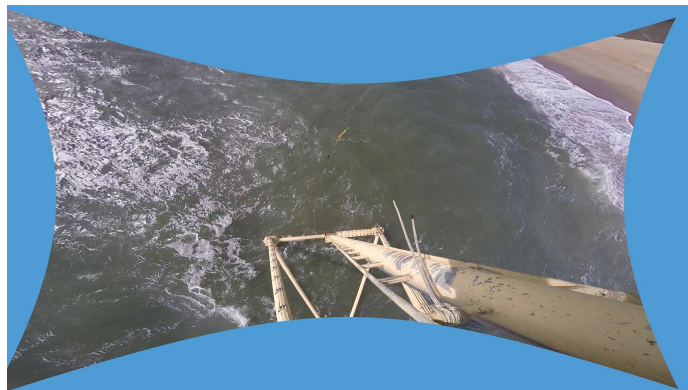
Data Processing- PIV

1. Split video into individual frames



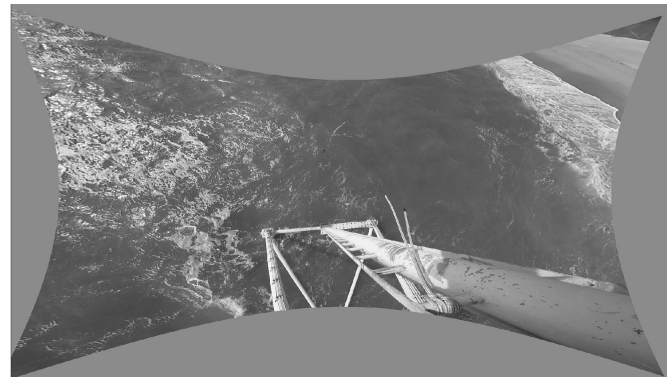
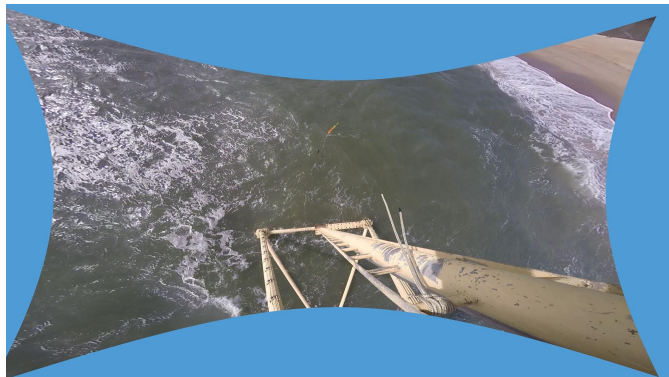
Data Processing- PIV

1. Split video into individual frames
2. **Undistort each frame for fisheye effect**



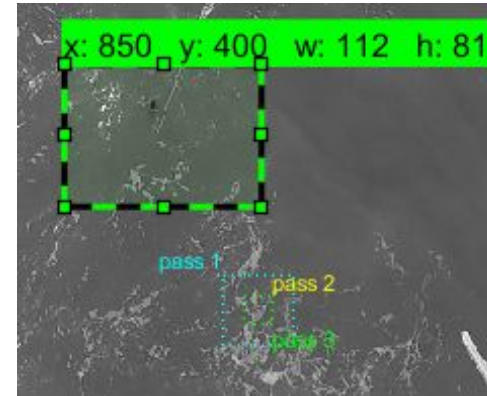
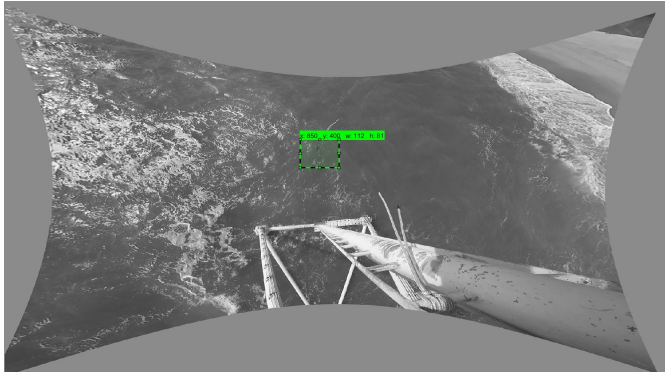
Data Processing- PIV

1. Split video into individual frames
2. Undistort each frame for fisheye effect
3. **Preprocess undistorted images to improve contrast of surface features**



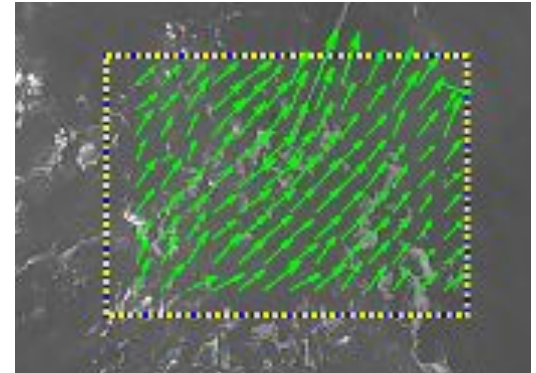
Data Processing- PIV

1. Split video into individual frames
2. Undistort each frame for fisheye effect
3. Preprocess undistorted images to improve contrast of surface features
4. **Set region of interest (ROI) and PIV settings**



Data Processing- PIV

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5. **Run PIV and export results [pixel displacements per frame]**

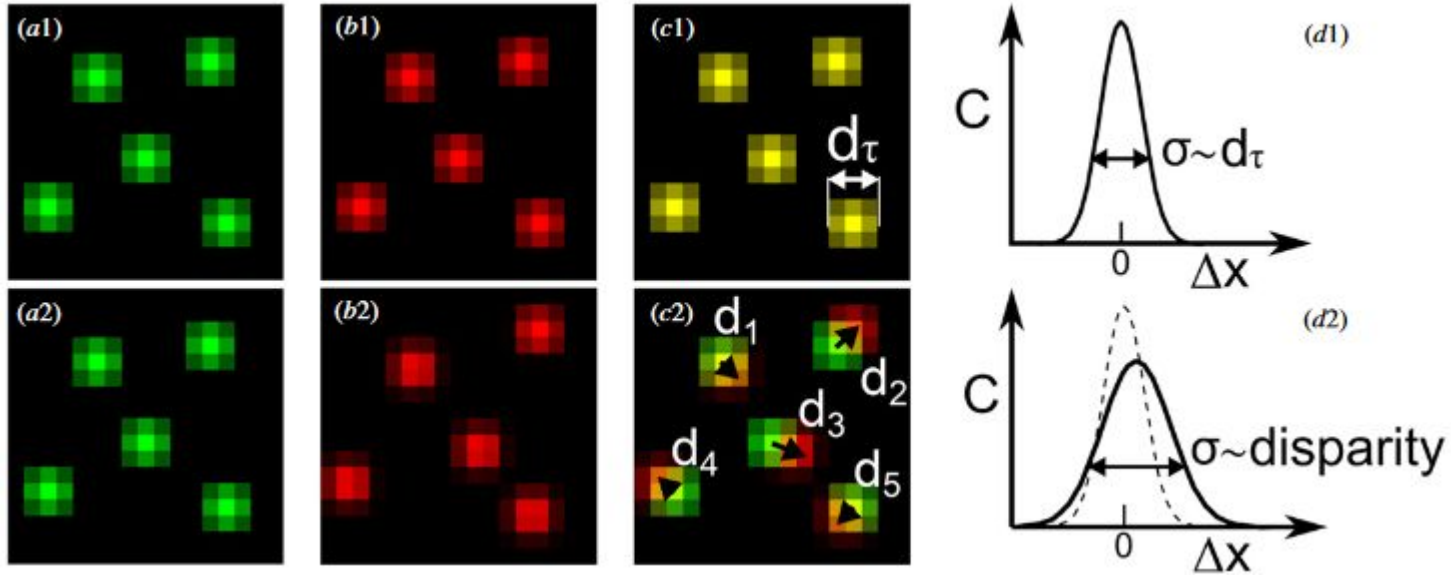


Data Processing- PIV

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6. **Quantify PIV uncertainty**

Data Processing- PIV

6. Quantify PIV uncertainty



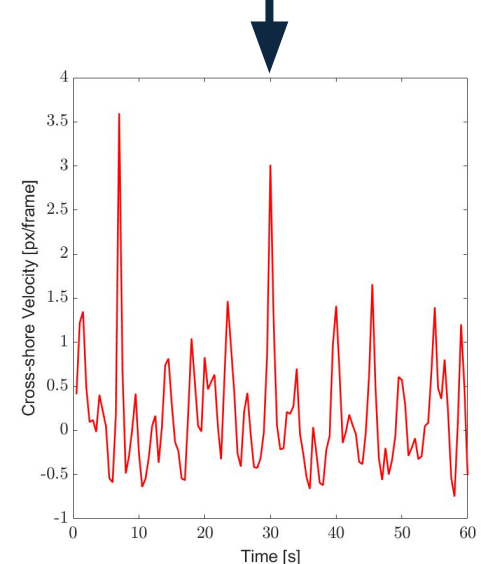
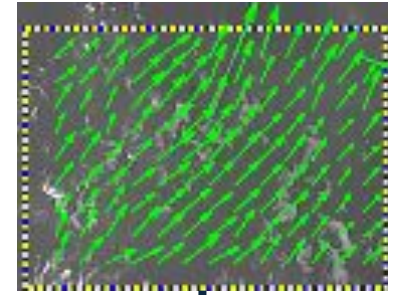
Sciacchitano, 2013

Data Processing– PIV

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4. Set region of interest (ROI) and PIV settings
5. Run PIV and export results [pixel displacements per frame]
6. Quantify PIV uncertainty
7. **Discard any PIV displacements with low correlation**

Data Processing- PIV

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7. Discard any PIV displacements with low correlation
8. **Take spatial mean of results**



Data Processing– PIV

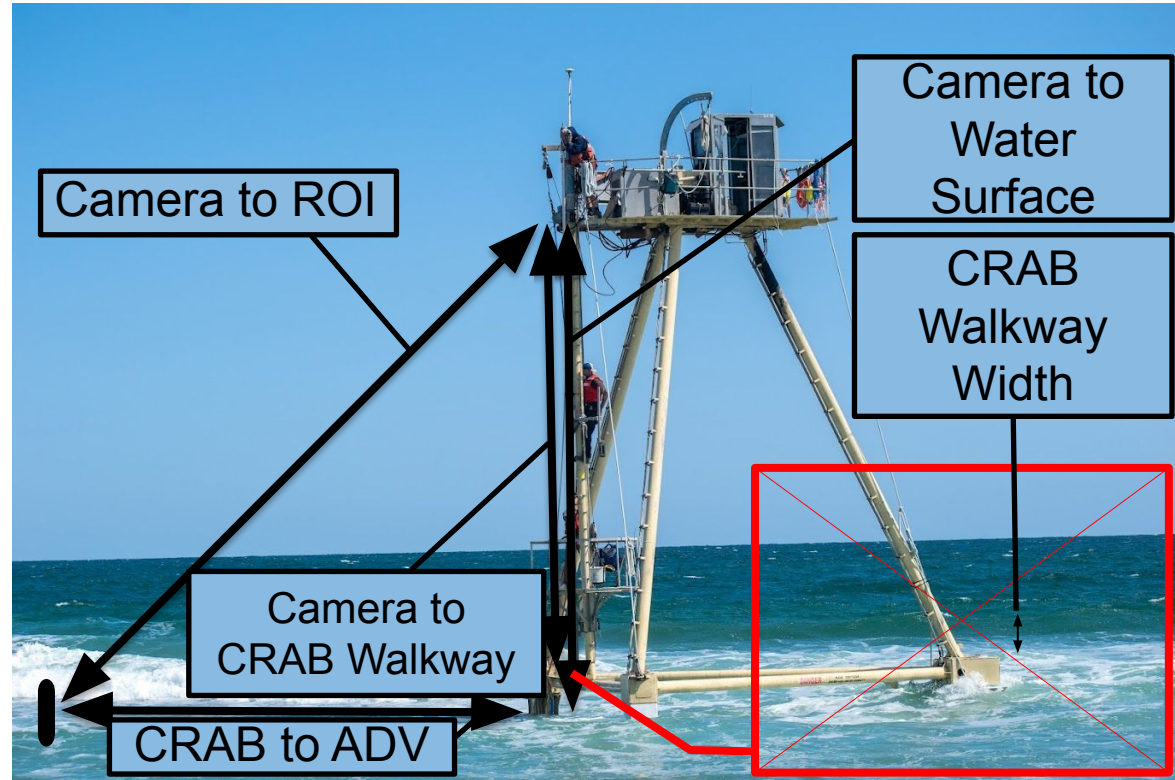
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6. Quantify PIV uncertainty
7. Discard any PIV displacements with low correlation
8. Take spatial mean of results
9. **Resampled PIV results to match ADV (30Hz $\square$ 2Hz)**

Data Processing– PIV

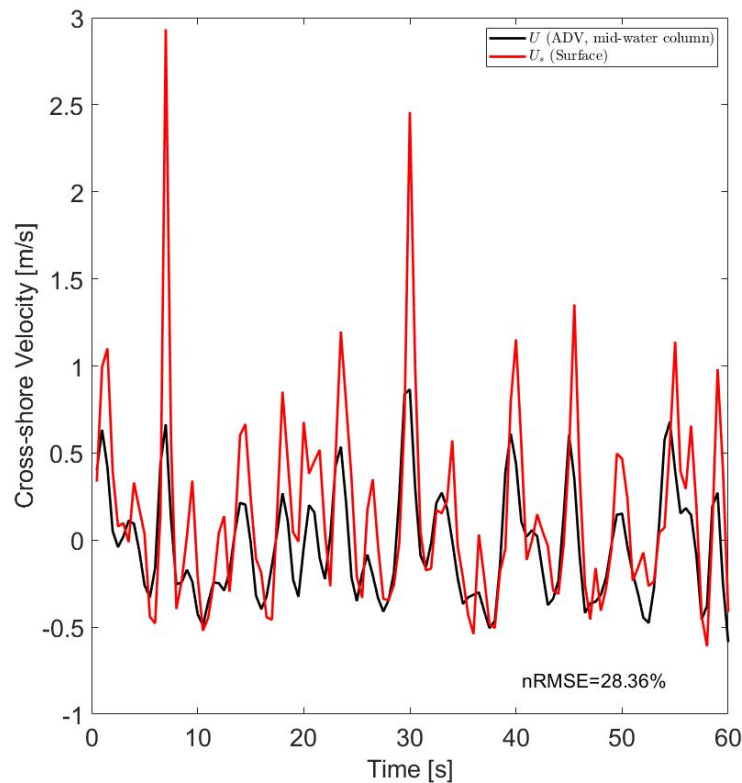
1. Split video into individual frames
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3. Preprocess undistorted images to improve contrast of surface features
4. Set region of interest (ROI) and PIV settings
5. Run PIV and export results [pixel displacements per frame]
6. Quantify PIV uncertainty
7. Discard any PIV displacements with low correlation
8. Take spatial mean of results
9. Resampled PIV results to match ADV (30Hz □ 2Hz)
- 10. Apply PIV calibration**

Data Processing- PIV

11. Apply PIV calibration



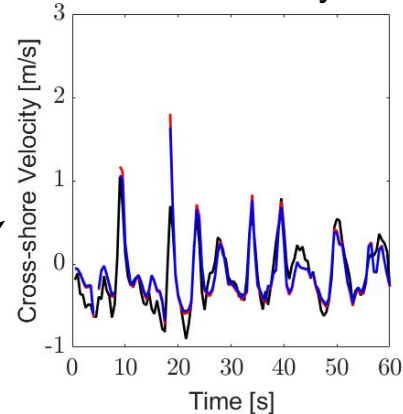
Data Processing- PIV output



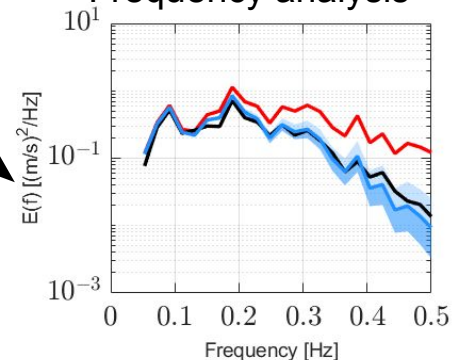
Data Processing- Overview



Time series analysis



Frequency analysis



Time domain analysis

$$U_c = U_s \frac{\cosh(k z_c)}{\cosh(k z_s)}$$

U_c = Corrected mid-water column velocity

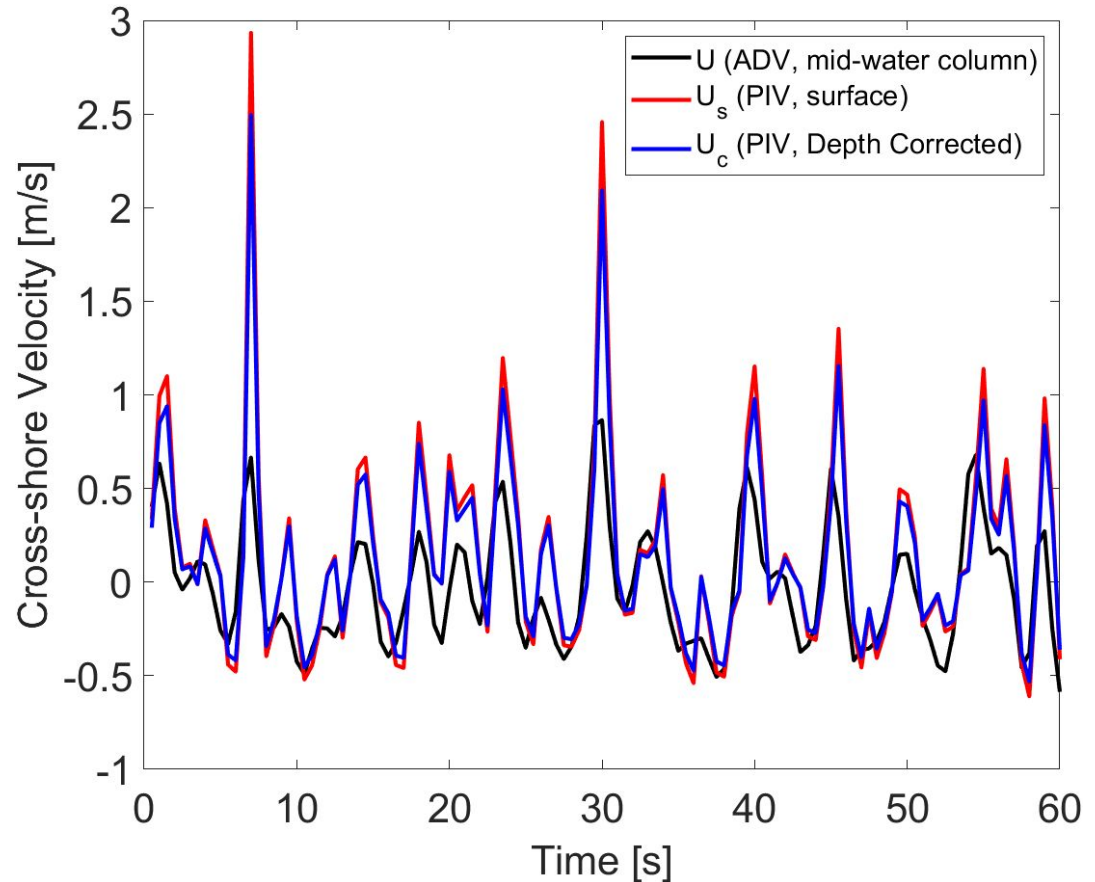
U_s = Surface velocity

k = Peak period wave number (derived from U_s)

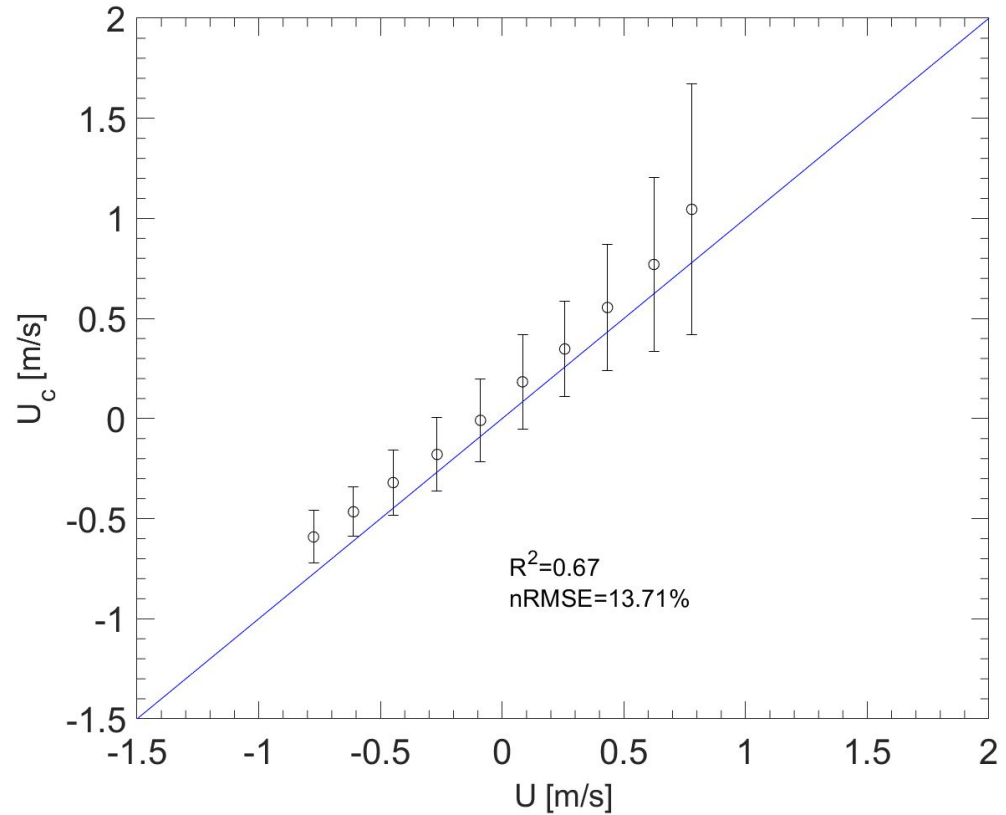
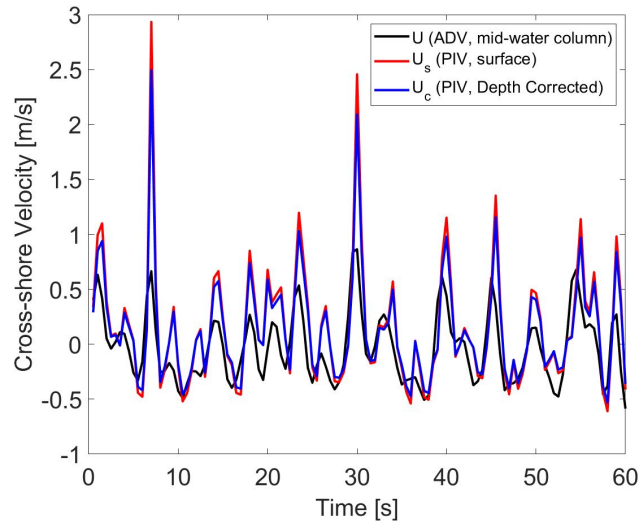
z_c = Depth of interest (height of ADV above bed)

z_s = Water depth

Results- Time domain analysis



Results– Time domain analysis



Frequency domain analysis

$$P_{U_c}(f) = P_{U_s}(f) \frac{\cosh^2(k(f) z_c)}{\cosh^2(k(f) z_s)}$$

$P_{U_c}(f)$ = Corrected mid-water column velocity power spectral density

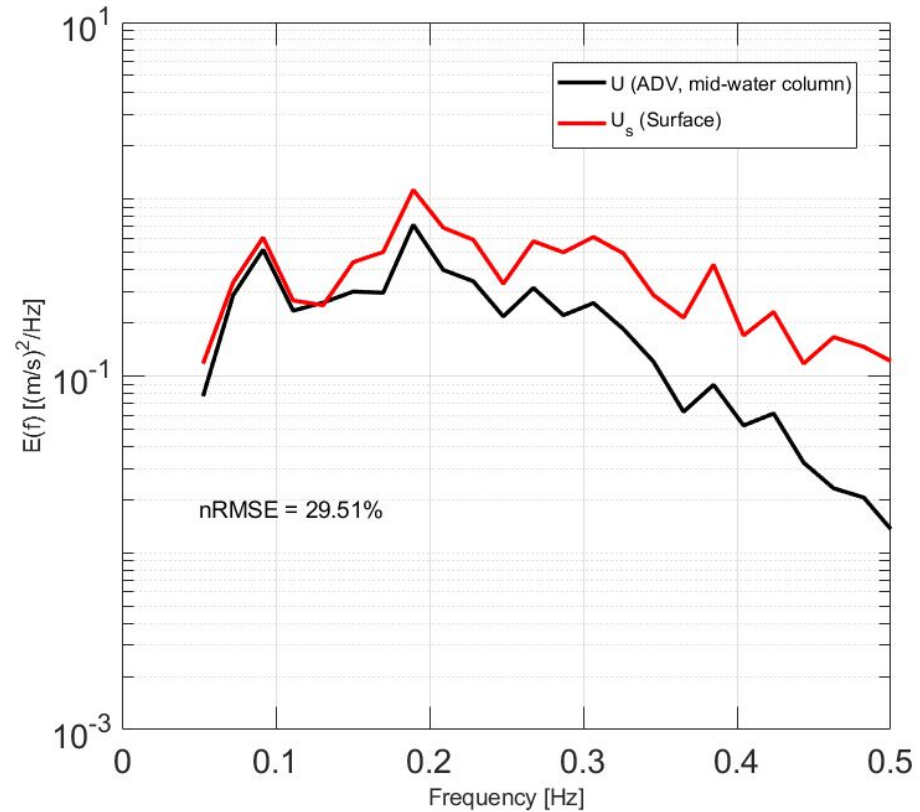
$P_{U_s}(f)$ = Surface velocity power spectral density

k = Wave number of each frequency bin (derived from U_s)

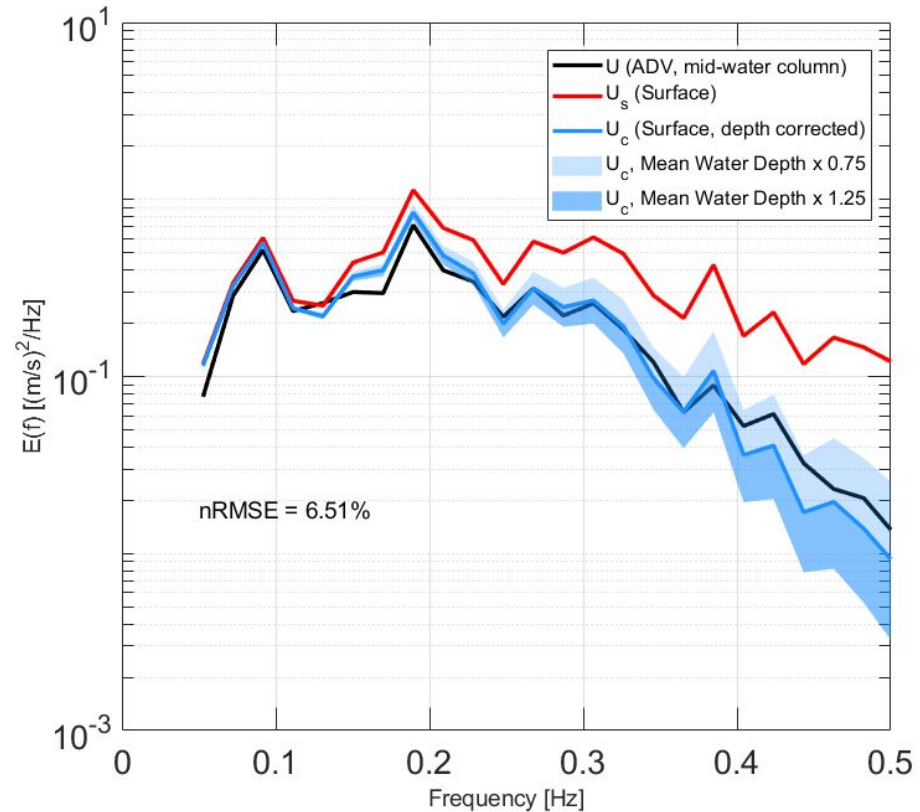
z_c = Depth of interest (height of ADV above bed)

z_s = Water depth

Results- Frequency domain analysis



Results- Frequency domain analysis



Non-linear time domain analysis

$$U_c = U_s \frac{\cosh(kz_c) + \frac{3}{8}k^2 \cosh(2kz_c)}{\cosh(kz_s) + \frac{3}{8}k^2 \cosh(2kz_s)}$$

U_c = Corrected mid-water column velocity

U_s = Surface velocity

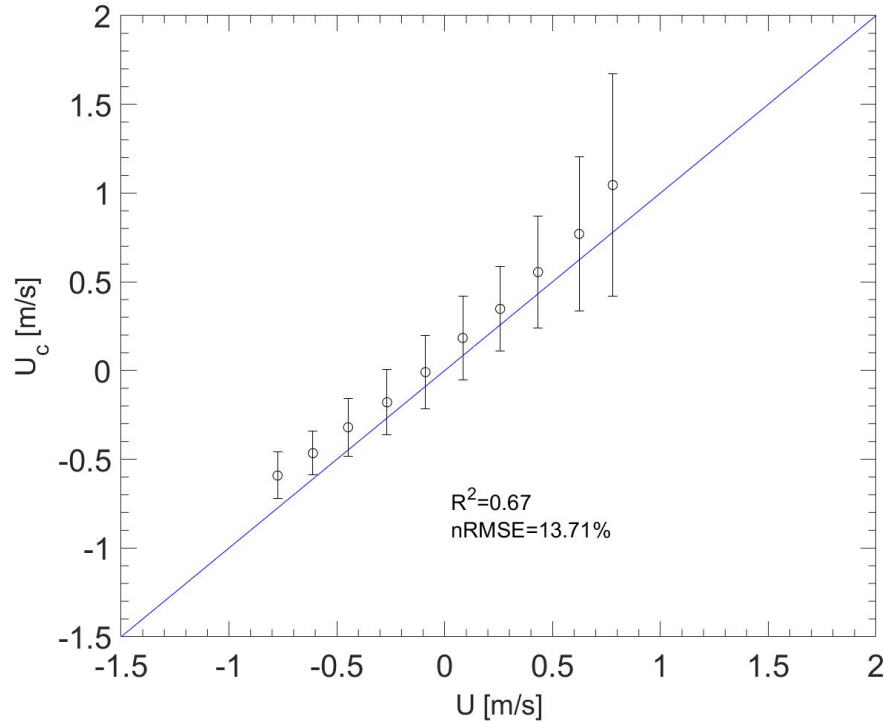
k = Peak period wave number (derived from U_s)

z_c = Depth of interest (height of ADV above bed)

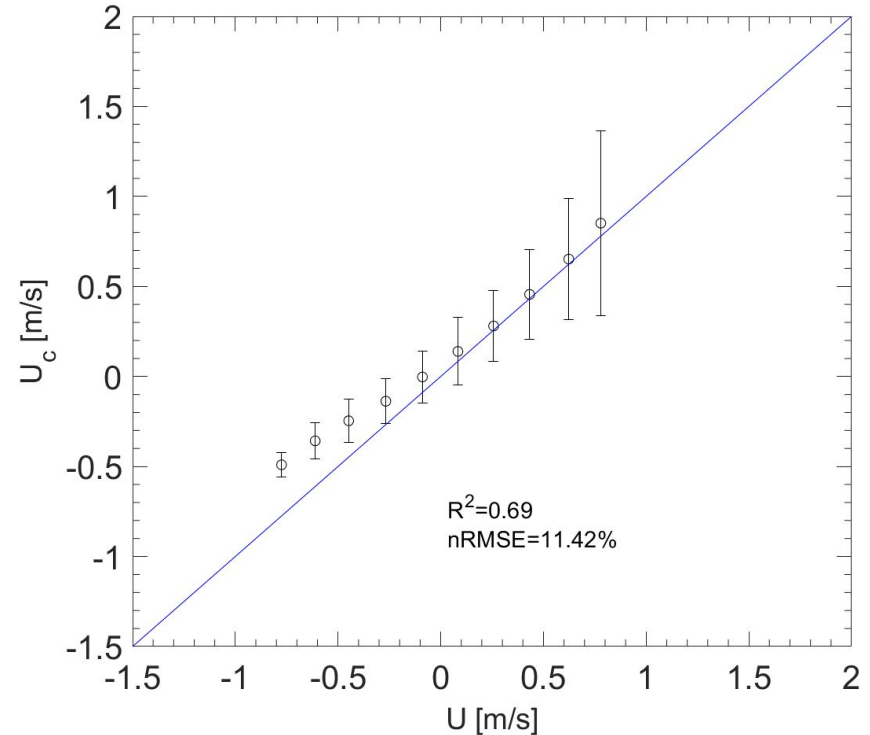
z_s = Water depth

Results | Non-linear time domain analysis

Linear (First order)



Non-linear (Second order)



Discussion

- A low-cost camera and linear-wave theory can be used to estimate orbital velocities with good accuracy in the surfzone
- Importance of water depth vs frequency information for LWT
- Advantages of both the time series and frequency domain methods
- This method can be used to increase spatiotemporal coverage of wave orbital velocities in the surfzone
 - Does not require in-situ instruments
 - Precise water depth not required
 - Can be implemented with existing cameras

Thank you!

Questions?



COASTAL &
HYDRAULICS
LABORATORY

